Actuator Selection for Tendon-Driven Humanoid Fingers

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Executive Summary

Robotic hands designed for human-centered environments often require a level of dexterity that cannot be achieved using simple grippers. As a result, humanoid hands frequently employ tendon-driven fingers, which allow multiple joints to be actuated while keeping the fingers lightweight and compact. In these systems, actuators are mounted remotely and transmit force to the finger joints through tendons and pulleys, making the relationship between actuator output and fingertip force dependent on the mechanical geometry of the finger. This technical brief introduces the basic structure of tendon-driven humanoid fingers and discusses factors that influence actuator selection for such mechanisms. A simplified example is presented to show how a known load applied at the fingertip of a three-joint curling finger can be used to estimate actuator torque requirements during early design stages. The example is intended to demonstrate the estimation process rather than provide an optimized design.

Introduction

With an increase in robots being deployed in human-based environments, humanoid hands are becoming increasingly integrated into robotic systems. The ability to manipulate objects of varying shapes and sizes allows for a wider range of tasks than the commonly used two-finger grippers.

Because all forces applied at the fingertip originate at the actuator, actuator selection plays a central role in determining the overall performance of tendon-driven humanoid fingers. As a result, improper actuator selection can lead to insufficient fingertip force capability or unnecessarily large and heavy systems. The following sections summarize tendon-driven finger

[1] This report was written entirely by me, and has not received any previous academic credit at this or any other institution.

architecture, actuator selection considerations, and present a sizing example for preliminary actuator selection.

Tendon-Driven Architectures

In tendon-driven systems, actuators do not act directly on the joints but instead apply force through flexible tendons routed throughout the finger. A typical tendon-driven finger consists of multiple revolute joints, often corresponding to the metacarpophalangeal (MCP), proximal interphalangeal (PIP), and distal interphalangeal (DIP) joints. Tendons are routed over pulleys or defined moment arms at each joint, allowing tendon tension to generate joint torque. In both cases, the torque generated at a joint depends on the applied tendon tension and the effective pulley radius at that joint [1].

To enable remote actuation, tendons are wound around a spool or drum attached to the actuator. The spool radius determines how much actuator torque is required to generate a given tendon tension and is therefore an important part of the overall transmission system.

Forces applied at the fingertip are transmitted through the finger joints, joint pulleys, tendons, and actuator spool before reaching the actuator. Changes in finger length, pulley radius, or spool radius can significantly alter the torque required from the actuator.

Actuator Selection Considerations

When selecting an actuator for a tendon-driven humanoid finger, consideration of how forces are transmitted through the finger and how external loads affect actuator demand are required. Unlike directly actuated joints, the actuator in a tendon-driven system does not experience joint torque directly. Instead, actuator torque is converted into tendon tension through the actuator spool and then into joint torque through pulleys which create moment arms at each joint [2]. As a result, actuator requirements depend strongly on the geometry of the finger and transmission system.

One of the primary considerations in actuator selection is the ability to generate enough torque to sustain expected fingertip loads. Just as actuator output torque is transmitted to the joints

through moment arms, so are external forces applied at the fingertip. These forces must be balanced by tendon tension, which in simplified static models is treated as uniform along the transmission path and ultimately limited by the torque capacity of the actuator. For example, longer finger segments and smaller pulley radii increase the required tendon tension, which in turn increases the torque demand placed on the actuator [3].

The geometry of the actuator spool also plays an important role in actuator selection. A larger spool radius increases the required torque to generate a given tendon tension, while a smaller spool radius reduces torque demand but increases the amount of actuator rotation needed to achieve a given joint displacement. This trade-off between torque and displacement must be considered when selecting an actuator, particularly in systems where range of motion and speed are constrained.

Additionally, efficiency losses within the transmission system further influence actuator requirements. Friction between tendons and pulleys, bending losses in the tendon routing, and inefficiencies in the actuator or gearbox reduce the effective force transmitted to the joints. For this reason, actuator sizing during preliminary design often includes additional margin to account for losses and uncertainties that are not captured by simplified static models.

Beyond torque capacity, actuator behavior under load is also an important consideration. Actuators with high gear reduction can provide large output torque but may resist backdriving, resulting in stiff finger behavior during interaction. Conversely, actuators that allow backdriving can enable passive compliance at the joints, improving safety and adaptability when interacting with external objects. This balance between torque capability and backdrivability is an important factor in the selection of actuators for tendon-driven humanoid fingers [2].

Actuator Sizing Example Calculation

A simplified sizing example is presented for a three-joint curling finger with a single flexor tendon routed over the DIP, PIP, and MCP joints (underactuated). An antagonistic tendon is assumed to provide return motion and pretension, while the net joint torque is generated by the difference between flexor and extensor tendon tensions. The external load is modeled as a

static force of $F = 10\text{N}$ applied at the fingertip. For the preliminary design example, the following assumptions are made: planar and static loading case, finger extended and the tip force acts approximately perpendicular to the finger (worst-case joint moments), and tendon tension is treated as uniform along the routing path (friction neglected).

The pulley radii are given as:

$$r_{DIP} = 0.45 \text{ cm} = 0.0045 \text{ m}, r_{PIP} = 0.65 \text{ cm} = 0.0065 \text{ m}, r_{MCP} = 0.8 \text{ cm} = 0.0080 \text{ m}$$

A. Joint torque requirement from a tip force

Let the finger segment lengths be L_1(proximal), L_2(middle), and L_3(distal). For a tip force F, the required joint torques are approximated as:

$$\tau_{DIP} \approx FL_3$$
$$\tau_{PIP} \approx F(L_2 + L_3)$$
$$\tau_{MCP} \approx F(L_1 + L_2 + L_3)$$

Using representative lengths $L_1 = 45$ mm, $L_2 = 25$ mm, $L_3 = 20$ mm:

$$\tau_{DIP} = 10\text{N}(0.020\text{m}) = 0.20 \text{ N} \cdot \text{m}$$
$$\tau_{PIP} = 10\text{N}(0.045\text{m}) = 0.45 \text{ N} \cdot \text{m}$$
$$\tau_{MCP} = 10\text{N}(0.090\text{m}) = 0.90\text{N} \cdot \text{m}$$

Figure 1 shows the general configuration of an underactuated 3 joint finger, with r1, r2, and r3 representing r_{MCP}, r_{PIP}, and r_{DIP} joints respectively, and L1, L2, and L3 representing the proximal, middle, and distal lengths [1].

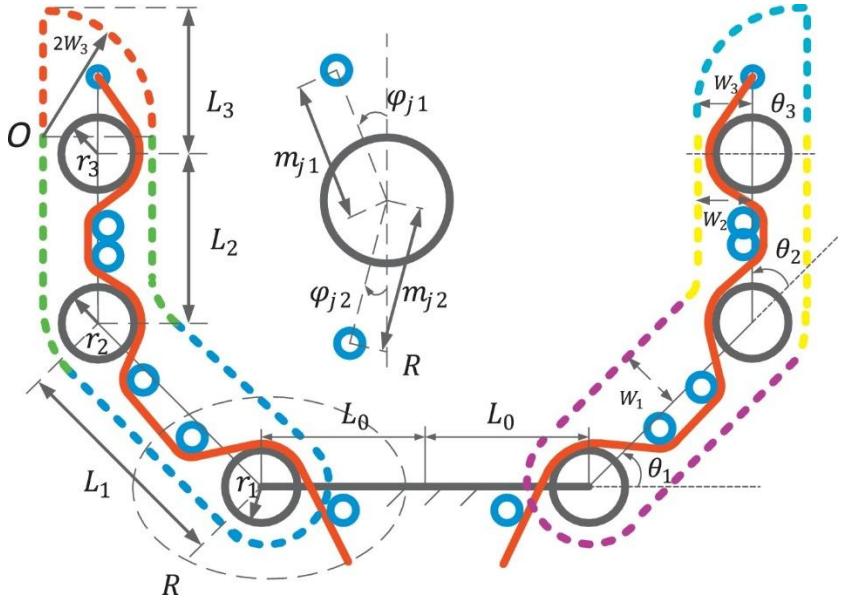

Fig. 1. Tendon routing and geometric parameters for a three-link underactuated finger [1]

B. Tendon tension requirement

For a tendon wrapped over a joint pulley, the joint torque is related to tendon tension by:

$$\tau_i = Tr_i \Rightarrow T_i = \frac{\tau_i}{r_i}$$

$$T_{DIP} = \frac{0.20\text{N} \cdot \text{m}}{0.0045\text{m}} = 44.4 \text{ N}$$

$$T_{PIP} = \frac{0.45\text{N} \cdot \text{m}}{0.0065\text{m}} = 69.2 \text{ N}$$

$$T_{MCP} = \frac{0.90\text{N} \cdot \text{m}}{0.0080\text{m}} = 112.5 \text{ N}$$

Because a single flexor tendon drives all three joints in this simplified model, the required net tendon tension (flexor minus extensor) is governed by the largest value:

$$T_{req} \approx 112.5 \text{ N}$$

C. Actuator torque requirement

If the tendon is wound on an actuator spool of radius r_s, the required actuator torque is:

$$\tau_{act} = T_{req} r_s$$

For a representative spool radius $r_s = 1.0 \text{ cm} = 0.010 \text{ m}$:

$$\tau_{act} = 112.5\text{N}(0.010\text{m}) = 1.125\text{N} \cdot \text{m}$$

This value represents an ideal static torque requirement and would typically be increased during preliminary design to account for frictional losses, routing inefficiencies, and uncertainty.

Conclusion

Tendon-driven humanoid fingers provide a practical approach for achieving dexterous manipulation while maintaining compact and lightweight finger structures. In these systems, actuator selection plays a critical role, as all fingertip forces must ultimately be generated by the actuator and transmitted through tendons, pulleys, and spools. As shown, actuator torque requirements depend strongly on finger geometry, joint pulley radii, and actuator spool dimensions rather than joint torque alone.

The simplified sizing example demonstrates how a known external fingertip load can be used to estimate tendon tension and actuator torque during preliminary design. While the example neglects friction, dynamic effects, and posture-dependent behavior, it illustrates the importance of considering transmission geometry when selecting actuators for underactuated tendon-driven fingers. Such first-order estimates are useful for early design stages, where actuator feasibility and system scale must be evaluated before more detailed modeling and testing are performed.

References

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